

Labdhi Shah

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Work Experience

Miscot System Private Ltd

December 2025–February 2026

*Intern as Software Developer
Mumbai*

- Developed and deployed **backend APIs using FastAPI** to integrate ML models and document-processing pipelines.
- Designed and developed a **hybrid retrieval-based chatbot** using embeddings, FAISS, JSON knowledge base and API-based LLM with similarity-threshold routing.
- Implemented **OCR pipelines** using PaddleOCR with DBNet for text detection and PARSeq for text recognition.
- Built and evaluated document text-extraction models using **DocTR, DBNet and PARSeq**, comparing performance and accuracy.
- Developed **AES encryption/decryption modules** supporting single and batch processing with secure padding mechanisms; researched image preprocessing techniques to improve model inputs.

Education

Thakur College of Engineering and Technology

July 2023 – Present

BTech in Artificial Intelligence and Data Science

Kapol Vidyanidhi International School

Secondary School

Technical Skills

Backend & Databases: FastAPI, Flask, REST APIs, PostgreSQL, MySQL, Supabase, Git

AI/ML: Machine Learning, NLP, LLMs, RAG, Scikit-learn, PyTorch, FAISS, OpenCV, OCR, Pandas, NumPy

AI Engineering: Agentic AI, MCP, Tool Calling, LLM Evaluation, Vector Search

Projects

L-Mart AI Loyalty Operations Agent

- Built and deployed an **agentic AI analyst** using Python, FastAPI, Pydantic, PostgreSQL/Supabase and Next.js, implementing a tool-calling architecture with **10 typed tools across 11 SQL-based business metrics**.
- Exposed the same tool registry through **Model Context Protocol (MCP)** alongside in-process dispatch, with a provider-neutral architecture supporting Claude and OpenAI-compatible gateways and durable, resumable investigations.
- Implemented **human-in-the-loop actions** with Supabase Auth, role-based authorization, row-level security, deterministic policy checks, hash-bound approvals, idempotent execution and append-only audit logging; added **121 automated tests**.

Knowledge Assistant — Grounded Document & Codebase Q&A;

- Built a full-stack **RAG assistant** using FastAPI, Next.js/TypeScript and PostgreSQL for Q&A; over PDF, DOCX and source-code files, with verified inline citations and private API connectivity.
- Engineered retrieval with **layout-aware PDF extraction, sentence-based chunking, section carry-forward, neighbour linking and traceable source offsets**, achieving **0.826 recall@3** on a 28-question gold set.
- A/B-tested chunking strategies and retrieval thresholds against the gold set, rejecting a recursive-separator chunker based on measured recall (**0.783 vs. 0.826**) and implementing deterministic evidence gating for unanswerable queries.

CarbonGrid — AI-Based Carbon Intelligence System

- Built a carbon footprint tracking platform using **Python, FastAPI/Flask and REST APIs** for sustainability analytics and net-zero planning.
- Engineered data-processing pipelines for **geospatial datasets, satellite imagery and LiDAR**, enabling environmental impact analysis.
- Integrated a **Gemini-based AI recommendation engine** and interactive dashboard to generate actionable sustainability insights.

Achievements & Leadership

2nd Place — IndiaNext Hackathon (National Level)

March 2026

- Secured **2nd position among 234 teams**; developed CarbonGrid, an AI-powered sustainability platform for net-zero planning in a 24-hour national hackathon.

Data Science Club (S4DS)

August 2024 – September 2025

- Led a **100+ member** student community and organized technical workshops and events with **300+ attendees**.